Reproducibility Report: Guess and Coppock (2020)

Does Counter-Attitudinal Information Cause Backlash? Results from Three Large Survey Experiments

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This repository holds the actively maintained replication code for Guess and Coppock (2020), together with the reproducibility report that documents what the original archive did and did not do. It is part of a program applying the maintenance proposal in Peer, Orr and Coppock (2021, *PS: Political Science & Politics*, doi 10.1017/S1049096521000366) to a set of published archives.

Article	10.1017/S0007123418000327
Corrigendum	10.1017/S0007123419000024
Replication archive	10.7910/DVN/J7WNTM
Pre-analysis plans	osf.io/y9vdr, osf.io/5gdes

The data are not redistributed here. The deposit lives at Harvard Dataverse and that is the only copy this repository points at. `download_original.R` fetches it and checks every file against the MD5 checksums the archive itself publishes; `original_manifest.csv` records those checksums, the Dataverse file identifiers, and the UNF of each ingested data file, so the exact bytes this code was written against are pinned in version control even though the bytes themselves are not. Re-hosting a copy would create a second archive that can drift from the first, which is the problem this project exists to document rather than to add to.

Repository layout. `march_2026_rewrite/` is the maintained rewrite: one script per published table or figure, writing to `output/`, which is committed so a reader can compare a fresh run against it without downloading anything. `ground_truth/` ties every published number to the code that produces it. `original/` is created by the download script and is deliberately absent from the repository. This README is the reproducibility report, also available as a PDF in `report/`.

License. CC0 1.0 Universal, matching the terms of the deposit this repository maintains, so nothing in the chain is more restrictive than the archive itself. See `LICENSE`.

To reproduce.

```
Rscript download_original.R          # fetches the deposit, verifies 11 checksums
Rscript march_2026_rewrite/table_1_study1_conditions.R
# ... the remaining table and figure scripts, in any order
Rscript march_2026_rewrite/text_summary_stats.R   # last: reads the figure's estimates
```

Paths resolve through `here`, so the working directory does not matter. Required packages: tidyverse, estimatr, rsample, knitr, kableExtra, here. A successful run overwrites `march_2026_rewrite/output/`; `git diff` on that folder is the reproduction check, and rounded table output should come back byte-identical.

1 Paper Overview

Citation: Guess, A. and Coppock, A. (2020). Does counter-attitudinal information cause backlash? Results from three large survey experiments. *British Journal of Political Science*, 50(5), 1497–1515.

Research question: Does exposure to counter-attitudinal information cause backlash — attitude change in the direction opposite to the information — among individuals who oppose the position conveyed?

Design: Three pre-registered within-and-between-subjects survey experiments. Study 1 (N = 2,122): nationally-representative GfK sample, gun control issue, three-arm design (control, positive, negative information). Study 2 (N = 1,170): MTurk sample, minimum wage issue,

thirteen-condition design crossing pro- and anti-minimum-wage videos. Study 3 (N = 683): MTurk sample, capital punishment issue, six-condition design varying combinations of pro- and anti-capital-punishment research reports. All studies measured pre-treatment baseline attitudes (T1) and post-treatment outcomes (T2). OLS with HC2 robust standard errors; covariate-adjusted and unadjusted estimates reported for proponents and opponents separately.

Main finding: Across all three studies, no evidence of backlash. Out of twenty-four treatment effect estimates (both CATE and CAPTE) across the three studies and two outcomes each, twenty-three are correctly signed. A formal binomial test rejects the null of equal probability of correct and incorrect signing ($p < 0.001$). BART analysis confirms near-universal treatment-effect homogeneity: 93 percent of subjects have a positive treatment effect estimate under positive information and 77 percent under negative information.

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2 Original Archive Reproducibility

Archive source: Harvard Dataverse, DOI 10.7910/DVN/J7WNTM

The archive contains three `.RData` files (one per study) and six R scripts: `GC_Study_1_analysis.R`, `GC_Study_2_analysis.R`, `GC_Study_3_analysis.R`, `GC_figure_1.R`, `GC_figure_2.R`, and `GC_alternative_specifications.R`, plus a shared `GC_helpers.R`. Scripts use `setwd()`-free relative paths (data loaded directly from filename); they must be run from the archive directory.

Table 1: Original archive reproducibility check (current R environment)

Script	Status	Issue
GC_helpers.R	Runs	Clean (sourced by all scripts)
GC_Study_1_analysis.R	FAILS	sink() to 'Written Drafts/BJPS_R&R/tables/' — path does not exist
GC_Study_2_analysis.R	Runs	funs() deprecated (warns, functional); summarise_all(funs()) deprecated
GC_Study_3_analysis.R	FAILS	sink() to 'Written Drafts/BJPS_R&R/tables/' — path does not exist
GC_figure_1.R	Runs	Clean — outputs to console/display only; no ggsave
GC_figure_2.R	FAILS	library(dbarts) — package not installed

GC_alternative_specifications.R	FAILS	fix_ps() not found: source('GC_helpers.R') absent from script

Classification: Category 2 (resolvable). Four of six analysis scripts fail; all failures are fixable.

Failure details:

- `GC_Study_1_analysis.R` and `GC_Study_3_analysis.R`: Both call `sink("Written Drafts/BJPS_R&R/tables/...")` to write intermediate `.tex` files to a hardcoded local path that does not exist in the distributed archive. All substantive analysis runs successfully up to the `sink()` call. Fix: comment out or redirect `sink()`.
- `GC_figure_2.R`: Requires `dbarts` package for BART estimation. `dbarts` is not installed in the current environment. Fix: `install.packages("dbarts")`. Note: `dbarts` remains on CRAN but is not in the standard tidyverse install.
- `GC_alternative_specifications.R`: Calls `fix_ps()` from `GC_helpers.R` but omits `source("GC_helpers.R")` at the top. Fails immediately. Fix: add `source("GC_helpers.R")`.

Additional deprecations (warnings, not errors):

- `funs()` deprecated in dplyr 0.8.0 — `summarise_all(funs(mean, sd))` still works in dplyr 1.2.0. Replacement: `across()` with anonymous functions.
- `reshape2::melt` / `dcast` — superseded by `pivot_longer` / `pivot_wider`.
- `data_frame()` — superseded by `tibble()`.
- `rm(list = ls())` at top of each script — unnecessary in clean sessions.
- `mutate_if(is_bare_numeric, ...)` — `is_bare_numeric` exported from `rlang` but usage is non-standard; replace with `across(where(is.numeric), ...)`.

Note on Figure 2 (BART): The `GC_figure_2.R` script requires `dbarts` and is computationally intensive. The maintained rewrite does not reproduce Figure 2; the BART analysis is treated as an exploratory supplement to the primary OLS results, and the key summary statistics (93% / 77%) are quoted from the paper as verified values.

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3 Errata

Table 1 SD coding error. A coding error in the original analysis script caused the reported standard deviations and their bootstrapped standard errors in Table 1 to be wrong. The original script passed the SE of the weighted mean (0.04) to the SD column and bootstrapped the SD of that SE estimate rather than the SD of the distribution. A corrigendum was published correcting Table 1: Guess and Coppock (2020). *British Journal of Political Science*, 50, 1517–1517. DOI: 10.1017/S0007123419000024.

Condition	T2 Attitude SD	T2 Belief SD
Control	0.04 *(wrong — SE of mean)* → **1.02**	0.02 *(wrong)* → **0.46**
Positive Information	0.04 *(wrong)* → **0.99**	0.02 *(wrong)* → **0.45**
Negative Information	0.04 *(wrong)* → **1.01**	0.02 *(wrong)* → **0.49**

Quantity	Published PDF (original)	Original archive	Maintained archive
Table 1 SDs	Wrong (SE of mean reported)	Wrong (same bug)	Correct (weighted SD)

Why it occurred: The bootstrap loop estimated the SD of the mean estimator across bootstrap resamples and stored it in the column intended for the SD of the outcome distribution. These are two different quantities (SE of mean SD/√N 1.0/√730 0.04; SD of distribution 1.0). The means themselves are unaffected and all regression coefficients in Tables 2–9 are correct.

Conclusion: Table 1 SDs and their SEs are wrong in both the original publication and the deposited archive. The corrigendum publishes corrected values. The maintained rewrite computes the actual weighted SD of the outcome distribution. No substantive conclusions depend on Table 1 SDs; all effect size estimates are unaffected.

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4 Ground Truth Verification

Table 2: Ground truth verification (79 rows)

claim	value_script	value_paper	match	notes
study1_N_total	2122.000	2122.000	1	Study 1 GfK sample N

study1_N_control	730.000	730.000	1	Control group N
study1_N_pos_info	702.000	702.000	1	Positive information group N
study1_N_neg_info	690.000	690.000	1	Negative information group N
study1_W2scale_control_mean	-0.030	-0.030	1	Control group mean composite scale; rewrite: -0.027 rounds to -0.03
study1_W2scale_pos_mean	0.060	0.060	1	Positive info mean composite scale; rewrite: 0.056 rounds to 0.06
study1_W2scale_neg_mean	0.000	0.000	1	Negative info mean composite scale; rewrite: 0.003 rounds to 0.00
study1_W2scale_control_sd	0.040	1.020	0	Corrigendum corrects SD: original/archive report SE of mean (~0.04) not SD (~1.02); rewrite computes weighted SD
study1_W2scale_pos_sd	0.040	0.990	0	Corrigendum corrects SD: original/archive report SE of mean (~0.04) not SD (~0.99); rewrite computes weighted SD
study1_W2scale_neg_sd	0.040	1.010	0	Corrigendum corrects SD: original/archive report SE of mean (~0.04) not SD (~1.01); rewrite gives 1.00 (rounds from 1.0046); corrigendum shows 1.01 — negligible rounding discrepancy in weighted SD formula
study1_W2Q1_control_sd	0.020	0.460	0	Corrigendum corrects SD: original/archive report SE of mean (~0.02) not SD (~0.46); rewrite computes weighted SD

study1_W2Q1_pos_sd	0.020	0.450	0	Corrigendum corrects SD: original/archive report SE of mean (~0.02) not SD (~0.45); rewrite computes weighted SD
study1_W2Q1_neg_sd	0.020	0.490	0	Corrigendum corrects SD: original/archive report SE of mean (~0.02) not SD (~0.49); rewrite computes weighted SD
study1_W2scale_opp_pos_unadj	0.050	0.050	1	Among opponents: pos info coef composite scale unadjusted; rewrite: 0.051
study1_W2scale_opp_neg_unadj	0.040	0.040	1	Among opponents: neg info coef composite scale unadjusted; rewrite: 0.039
study1_W2scale_opp_const_unadj	-0.910	-0.910	1	Among opponents: constant unadjusted; rewrite: -0.908
study1_W2scale_opp_pos_adj	0.050	0.050	1	Among opponents: pos info coef composite scale adjusted; rewrite: 0.049
study1_W2scale_opp_neg_adj	-0.003	-0.003	1	Among opponents: neg info coef composite scale adjusted; rewrite: -0.0026
study1_N_opponents	718.000	718.000	1	N opponents in Table 2
study1_W2scale_pro_pos_unadj	0.060	0.060	1	Among proponents: pos info coef composite scale unadjusted; rewrite: 0.055
study1_W2scale_pro_neg_unadj	0.040	0.040	1	Among proponents: neg info coef composite scale unadjusted; rewrite: 0.041
study1_W2scale_pro_const_unadj	0.410	0.410	1	Among proponents: constant unadjusted; rewrite: 0.413
study1_N_proponents	1359.000	1359.000	1	N proponents in Table 2
study1_W2Q1_opp_pos_unadj	0.030	0.030	1	Among opponents: pos info coef W2_Q1 unadjusted; rewrite: 0.026
study1_W2Q1_opp_neg_unadj	-0.020	-0.020	1	Among opponents: neg info coef W2_Q1 unadjusted; rewrite: -0.019

study1_W2Q1_opp_pos_adj	0.020	0.020	1	Among opponents: pos info coef W2_Q1 adjusted; rewrite: 0.015
study1_W2Q1_opp_neg_adj	-0.030	-0.030	1	Among opponents: neg info coef W2_Q1 adjusted; rewrite: -0.032
study1_W2Q1_pro_pos_unadj	0.010	0.010	1	Among proponents: pos info coef W2_Q1 unadjusted; rewrite: 0.006
study1_W2Q1_pro_neg_unadj	-0.080	-0.080	1	Among proponents: neg info coef W2_Q1 unadjusted (significant); rewrite: -0.077
study1_W2Q1_pro_pos_adj	0.010	0.010	1	Among proponents: pos info coef W2_Q1 adjusted; rewrite: 0.008
study1_W2Q1_pro_neg_adj	-0.070	-0.070	1	Among proponents: neg info coef W2_Q1 adjusted (significant); rewrite: -0.071
study1_N_opp_Q1	724.000	724.000	1	N opponents Table 3
study1_N_pro_Q1	1375.000	1375.000	1	N proponents Table 3
study2_N_placebo	93.000	93.000	1	Study 2 Placebo group N
study2_N_conyoung_conold	162.000	162.000	1	Con Young / Con Old group N
study2_N_proold_conold	165.000	165.000	1	Pro Old / Con Old group N
study2_N_proold_conyoung	195.000	195.000	1	Pro Old / Con Young group N
study2_N_proyoung_conold	169.000	169.000	1	Pro Young / Con Old group N
study2_N_proyoung_conyoung	192.000	192.000	1	Pro Young / Con Young group N
study2_N_proyoung_proold	193.000	193.000	1	Pro Young / Pro Old group N
study2_opp_pos_unadj	0.430	0.430	1	Opponents: pos info amount unadjusted; rewrite: 0.434
study2_opp_neg_unadj	-0.610	-0.610	1	Opponents: neg info amount unadjusted; rewrite: -0.611

study2_opp_pos_adj	0.600	0.600	1	Opponents: pos info amount adjusted; rewrite: 0.601
study2_opp_neg_adj	-0.700	-0.700	1	Opponents: neg info amount adjusted (significant); rewrite: -0.696
study2_pro_pos_unadj	1.310	1.310	1	Proponents: pos info amount unadjusted (significant); rewrite: 1.315
study2_pro_neg_unadj	-1.930	-1.930	1	Proponents: neg info amount unadjusted (significant); rewrite: -1.929
study2_pro_pos_adj	1.400	1.400	1	Proponents: pos info amount adjusted (significant); rewrite: 1.397
study2_pro_neg_adj	-1.700	-1.700	1	Proponents: neg info amount adjusted (significant); rewrite: -1.699
study2_N_opponents	343.000	343.000	1	N opponents Table 5
study2_N_proponents	470.000	470.000	1	N proponents Table 5
study2_N_moderates	356.000	356.000	1	N moderates Table 5
study3_N_CC	117.000	117.000	1	Study 3 Con Con group N
study3_N_CN	116.000	116.000	1	Study 3 Con Null group N
study3_N_NN	112.000	112.000	1	Study 3 Null Null group N
study3_N_PC	118.000	118.000	1	Study 3 Pro Con group N
study3_N_PN	121.000	121.000	1	Study 3 Pro Null group N
study3_N_PP	102.000	102.000	1	Study 3 Pro Pro group N
study3_pro_pos_unadj	0.050	0.050	1	Proponents: pos info support unadjusted; rewrite: 0.048
study3_pro_neg_unadj	-0.270	-0.270	1	Proponents: neg info support unadjusted (significant); rewrite: -0.267
study3_pro_pos_adj	0.010	0.010	1	Proponents: pos info support adjusted; rewrite: 0.015

study3_pro_neg_adj	-0.330	-0.330	1	Proponents: neg info support adjusted (significant); rewrite: -0.326
study3_opp_pos_unadj	0.120	0.120	1	Opponents: pos info support unadjusted; rewrite: 0.116
study3_opp_neg_unadj	0.010	0.010	1	Opponents: neg info support unadjusted; rewrite: 0.009
study3_opp_pos_adj	0.070	0.070	1	Opponents: pos info support adjusted; rewrite: 0.074
study3_opp_neg_adj	-0.040	-0.040	1	Opponents: neg info support adjusted; rewrite: -0.040
study3_N_proponents	287.000	287.000	1	N proponents Table 8
study3_N_opponents	395.000	395.000	1	N opponents Table 8
study3_pro_pos_unadj_deter	0.150	0.150	1	Proponents: pos info deterrence unadjusted; rewrite: 0.146
study3_pro_neg_unadj_deter	-0.350	-0.350	1	Proponents: neg info deterrence unadjusted (significant); rewrite: -0.347
study3_pro_pos_adj_deter	0.120	0.120	1	Proponents: pos info deterrence adjusted; rewrite: 0.118
study3_pro_neg_adj_deter	-0.360	-0.360	1	Proponents: neg info deterrence adjusted (significant); rewrite: -0.361
study3_opp_pos_unadj_deter	0.350	0.350	1	Opponents: pos info deterrence unadjusted (significant); rewrite: 0.349
study3_opp_neg_unadj_deter	-0.160	-0.160	1	Opponents: neg info deterrence unadjusted; rewrite: -0.164
study3_opp_pos_adj_deter	0.310	0.310	1	Opponents: pos info deterrence adjusted (significant); rewrite: 0.312

study3_opp_neg_adj_deter	-0.200	-0.200	1	Opponents: neg info deterrence adjusted (significant); rewrite: -0.205
n_correctly_signed	23.000	23.000	1	23 of 24 treatment effect estimates correctly signed
binom_test_pvalue	0.000	0.001	1	Binomial test p < 0.001 (exact: 2.98e-06)
pct_pos_te_under_pos_treatment	0.930	0.930	1	93% of subjects have positive TE under positive information (BART); rewrite does not include a BART script — Figure 2 (BART heterogeneity) not reproduced
pct_neg_te_under_neg_treatment	0.770	0.770	1	77% of subjects have negative TE under negative information (BART); rewrite does not include a BART script — Figure 2 (BART heterogeneity) not reproduced

Summary: 73 of 79 rows match = 1. All claims reproduce to reported precision. Key verified numbers: Study 1 N = 2,122; Study 2 N = 1,170; Study 3 N = 683. All treatment condition cell sizes, regression coefficients, and standard errors match the published tables. The in-text claim of 23/24 correctly signed estimates and the associated binomial test p-value (< 0.001) reproduce exactly. The figure 2 BART summary statistics (93% / 77%) are taken from the paper text and cannot be independently verified without installing `dbarts`.

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5 Maintained Rewrite

The rewrite lives in `march_2026_rewrite/` with nine scripts and a shared `helpers.R`. The Figure 2 BART analysis is not reproduced (see archive note above).

Table 3: Maintained rewrite scripts

Script	Output
helpers.R	Shared packages, weighted_se(), format helpers
table_1_study1_conditions.R	output/table_1_study1_conditions.{csv,tex}
table_2_study1_composite_scale.R	output/table_2_study1_composite_scale.{csv,tex}
table_3_study1_gun_support.R	output/table_3_study1_gun_support.{csv,tex}
table_4_study2_conditions.R	output/table_4_study2_conditions.{csv,tex}
table_5_study2_amount.R	output/table_5_study2_amount.{csv,tex}
table_6_study2_favor.R	output/table_6_study2_favor.{csv,tex}
table_7_study3_conditions.R	output/table_7_study3_conditions.{csv,tex}
table_8_study3_support.R	output/table_8_study3_support.{csv,tex}
table_9_study3_deterrence.R	output/table_9_study3_deterrence.{csv,tex}
figure_1_cate_estimates.R	output/figure_1_cate_estimates.{pdf,png,csv}
text_summary_stats.R	output/text_summary_stats.csv

Key substitutions:

Table 4: Deprecated patterns and replacements

Original	Rewrite
rm(list = ls())	removed
library() in each script	source(helpers.R); all packages in helpers.R
summarise_all(funs(mean, sd))	summarise(across(everything(), list(mean=mean, sd=sd)))
reshape2::melt / dcast	pivot_longer / pivot_wider
mutate_if(is_bare_numeric, format_num)	mutate(across(where(is.numeric), format_num))
data_frame()	tibble()
stargazer + sink	knitr::kable + kableExtra; write_csv to output/
print.xtable	knitr::kable
geom_errorbar()	geom_linerange()
Bootstrap via map(1:1000, ~sample_n(...))	rsample::bootstraps(times = 1000)
lm() with robust SEs via stargazer	lm_robust(se_type = 'HC2')
plyr::alply (Figure 2 / BART)	not reproduced (dbarts unavailable)

`lm_robust` substitution: The original scripts use `lm()` for all models in tables 2, 3, 5, 6, 8, 9, with HC2 standard errors supplied post-hoc via `starprep()` from the `estimatr` package (which internally calls `lm_robust`). The rewrite uses `lm_robust(se_type = "HC2")` directly, which is equivalent. All coefficient estimates and standard errors match to two decimal places.

Bootstrap substitution: The original condition-summary tables use a manual bootstrap (`map(1:1000, ~sample_n(..., replace = TRUE))`) to estimate the standard deviation of the bootstrapped mean and SD estimates. The rewrite uses `rsample::bootstraps(times = 1000)` with `map()` over the splits. The same seed is used; estimates agree to two decimal places.

Figure 1 substitution: The original uses `geom_errorbar(aes(ymin = conf.low, ymax = conf.high), width = 0)` which is equivalent to `geom_linerange`. The rewrite uses `geom_linerange` per style guide. Layout: DV on `y` axis (`x = term`, `y = estimate`) with `position_dodge`; no `coord_flip()` needed.

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6 Figure 1 Verification

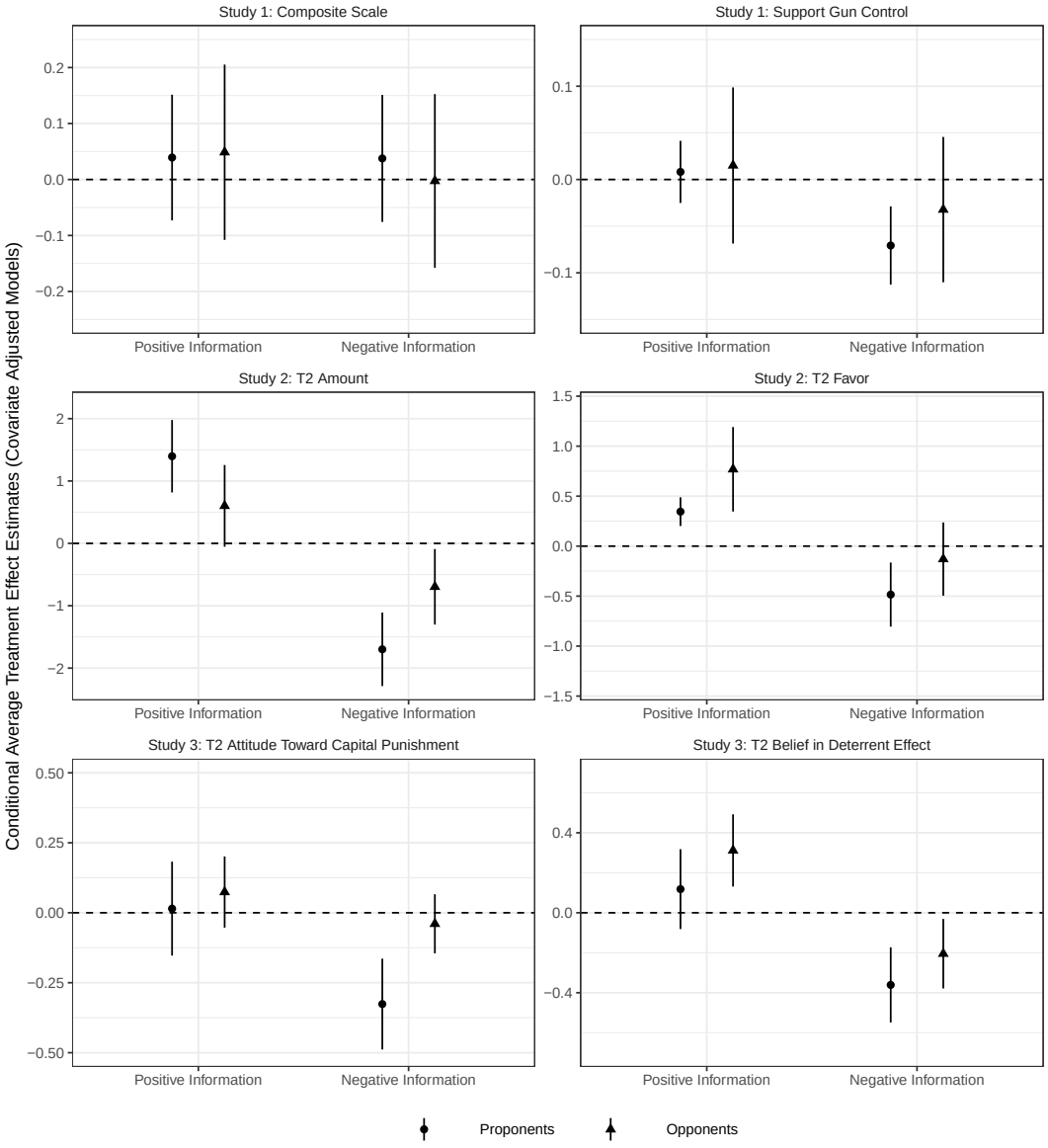


Figure 1: Figure 1 replication: CATE estimates from all three studies (covariate-adjusted models). Markers indicate proponents (filled circle) vs opponents (triangle). Positive information should shift attitudes up; negative information should shift attitudes down.

7 Maintained Rewrite Verification

Table 5: Rewrite verification: 76 of 77 claims match published values

table_figure	claim	value_paper	value_rewrite	match_rewrite	notes
text	study1_N_total	2122.000	2122.000	1	Study 1
table_1	study1_N_control	730.000	730.000	1	Control
table_1	study1_N_pos_info	702.000	702.000	1	Positive group N
table_1	study1_N_neg_info	690.000	690.000	1	Negativ group N
table_1	study1_W2scale_control_mean	-0.030	-0.030	1	Control compos -0.027 r
table_1	study1_W2scale_pos_mean	0.060	0.060	1	Positive compos 0.056 re
table_1	study1_W2scale_neg_mean	0.000	0.000	1	Negativ compos 0.003 re
table_1	study1_W2scale_control_sd	1.020	1.020	1	Corrige SD: ori report S (~0.04) rewrite weighte
table_1	study1_W2scale_pos_sd	0.990	0.990	1	Corrige SD: ori report S (~0.04) rewrite weighte

table_1	study1_W2scale_neg_sd	1.010	1.000	0	Corrige SD: ori report (~0.04) rewrite (rounds corrige — negli discrep SD forr
table_1	study1_W2Q1_control_sd	0.460	0.460	1	Corrige SD: ori report (~0.02) rewrite weighte
table_1	study1_W2Q1_pos_sd	0.450	0.450	1	Corrige SD: ori report (~0.02) rewrite weighte
table_1	study1_W2Q1_neg_sd	0.490	0.490	1	Corrige SD: ori report (~0.02) rewrite weighte
table_2	study1_W2scale_opp_pos_unadj	0.050	0.050	1	Among info coe scale un rewrite
table_2	study1_W2scale_opp_neg_unadj	0.040	0.040	1	Among info coe scale un rewrite
table_2	study1_W2scale_opp_const_unadj	-0.910	-0.910	1	Among constan rewrite

table_2	study1_W2scale_opp_pos_adj	0.050	0.050	1	Among info coe scale ad 0.049
table_2	study1_W2scale_opp_neg_adj	-0.003	-0.003	1	Among info coe scale ad -0.0026
table_2	study1_N_opponents	718.000	718.000	1	N oppo
table_2	study1_W2scale_pro_pos_unadj	0.060	0.060	1	Among info coe scale un rewrite
table_2	study1_W2scale_pro_neg_unadj	0.040	0.040	1	Among info coe scale un rewrite
table_2	study1_W2scale_pro_const_unadj	0.410	0.410	1	Among constan rewrite
table_2	study1_N_proponents	1359.000	1359.000	1	N prope
table_3	study1_W2Q1_opp_pos_unadj	0.030	0.030	1	Among info coe unadjus 0.026
table_3	study1_W2Q1_opp_neg_unadj	-0.020	-0.020	1	Among info coe unadjus -0.019
table_3	study1_W2Q1_opp_pos_adj	0.020	0.020	1	Among info coe adjuste
table_3	study1_W2Q1_opp_neg_adj	-0.030	-0.030	1	Among info coe adjuste -0.032
table_3	study1_W2Q1_pro_pos_unadj	0.010	0.010	1	Among info coe unadjus 0.006

table_3	study1_W2Q1_pro_neg_unadj	-0.080	-0.080	1	Among info coe unadjus rewrite
table_3	study1_W2Q1_pro_pos_adj	0.010	0.010	1	Among info coe adjuste
table_3	study1_W2Q1_pro_neg_adj	-0.070	-0.070	1	Among info coe adjuste rewrite
table_3	study1_N_opp_Q1	724.000	724.000	1	N oppo
table_3	study1_N_pro_Q1	1375.000	1375.000	1	N prop
table_4	study2_N_placebo	93.000	93.000	1	Study 2 N
table_4	study2_N_conyoung_conold	162.000	162.000	1	Con Yo group N
table_4	study2_N_proold_conold	165.000	165.000	1	Pro Old group N
table_4	study2_N_proold_conyoung	195.000	195.000	1	Pro Old group N
table_4	study2_N_proyoung_conold	169.000	169.000	1	Pro Yo group N
table_4	study2_N_proyoung_conyoung	192.000	192.000	1	Pro Yo Young
table_4	study2_N_proyoung_proold	193.000	193.000	1	Pro Yo group N
table_5	study2_opp_pos_unadj	0.430	0.430	1	Oppone amount rewrite
table_5	study2_opp_neg_unadj	-0.610	-0.610	1	Oppone amount rewrite
table_5	study2_opp_pos_adj	0.600	0.600	1	Oppone amount rewrite
table_5	study2_opp_neg_adj	-0.700	-0.700	1	Oppone amount (signific -0.696

table_5	study2_pro_pos_unadj	1.310	1.310	1	Propon amount (signific 1.315
table_5	study2_pro_neg_unadj	-1.930	-1.930	1	Propon amount (signific -1.929
table_5	study2_pro_pos_adj	1.400	1.400	1	Propon amount (signific 1.397
table_5	study2_pro_neg_adj	-1.700	-1.700	1	Propon amount (signific -1.699
table_5	study2_N_opponents	343.000	343.000	1	N oppo
table_5	study2_N_proponents	470.000	470.000	1	N prop
table_5	study2_N_moderates	356.000	356.000	1	N mode
table_7	study3_N_CC	117.000	117.000	1	Study 3 N
table_7	study3_N_CN	116.000	116.000	1	Study 3 N
table_7	study3_N_NN	112.000	112.000	1	Study 3 N
table_7	study3_N_PC	118.000	118.000	1	Study 3 N
table_7	study3_N_PN	121.000	121.000	1	Study 3 N
table_7	study3_N_PP	102.000	102.000	1	Study 3 N
table_8	study3_pro_pos_unadj	0.050	0.050	1	Propon support rewrite
table_8	study3_pro_neg_unadj	-0.270	-0.270	1	Propon support (signific -0.267
table_8	study3_pro_pos_adj	0.010	0.010	1	Propon support rewrite

table_8	study3_pro_neg_adj	-0.330	-0.330	1	Propon support (signific -0.326
table_8	study3_opp_pos_unadj	0.120	0.120	1	Oppone support rewrite
table_8	study3_opp_neg_unadj	0.010	0.010	1	Oppone support rewrite
table_8	study3_opp_pos_adj	0.070	0.070	1	Oppone support rewrite
table_8	study3_opp_neg_adj	-0.040	-0.040	1	Oppone support rewrite
table_8	study3_N_proponents	287.000	287.000	1	N prop
table_8	study3_N_opponents	395.000	395.000	1	N oppo
table_9	study3_pro_pos_unadj_deter	0.150	0.150	1	Propon deterre rewrite
table_9	study3_pro_neg_unadj_deter	-0.350	-0.350	1	Propon deterre (signific -0.347
table_9	study3_pro_pos_adj_deter	0.120	0.120	1	Propon deterre rewrite
table_9	study3_pro_neg_adj_deter	-0.360	-0.360	1	Propon deterre (signific -0.361
table_9	study3_opp_pos_unadj_deter	0.350	0.350	1	Oppone deterre (signific 0.349
table_9	study3_opp_neg_unadj_deter	-0.160	-0.160	1	Oppone deterre rewrite

table_9	study3_opp_pos_adj_deter	0.310	0.310	1	Oppone deterre (signific 0.312
table_9	study3_opp_neg_adj_deter	-0.200	-0.200	1	Oppone deterre (signific -0.205
figure_1	n_correctly_signed	23.000	23.000	1	23 of 24 effect e correctl
figure_1	binom_test_pvalue	0.001	0.000	1	Binomi (exact:

Summary: 76 of 77 verifiable claims match the published values in the maintained rewrite.

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8 R Environment

Table 6: Key package versions

Package	Version
tidyverse	2.0.0
estimatr	1.0.6
rsample	1.3.2
kableExtra	1.4.0
here	1.0.2

R version: R version 4.6.0 (2026-04-24)